CURRICULUM VITAE

Kagan Simsek
(also publishes as Kaan Şimşek)
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CONTACT INFORMATION

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PROFESSIONAL SUMMARY

Theoretical particle physicist with over seven years of research experience in precision phenomenology, collider physics, and physics beyond the Standard Model. Expertise spans perturbative QCD and electroweak theory, the Standard Model Effective Field Theory (SMEFT), hadron phenomenology, and higher-order radiative corrections. Author of 12 peer-reviewed journal publications and 3 additional scholarly works (including 1 preprint, 1 community white paper, and 1 conference proceeding) with 379 citations and an h-index of 9 (INSPIRE), a publication record characterized by collaborators as exceptionally strong for a researcher at this career stage.

Contributed SMEFT operator constraint projections to the Electron-Ion Collider (EIC) Yellow Report, a community white paper that helped shape the scientific case for the EIC, a major U.S. Department of Energy facility under construction at Brookhaven National Laboratory. Subsequent work established the EIC, the Large Hadron-electron Collider (LHeC), and the Future Circular Collider (FCC) as powerful probes of physics beyond the Standard Model, with direct relevance to the scientific programs of these facilities. Current research at Kennesaw State University focuses on higher-order QCD corrections to top-quark production and SMEFT interpretations at the Large Hadron Collider, supported by NSF and university funding.

Dedicated educator with over ten years of teaching and mentoring experience across 34 semesters at Middle East Technical University (Ankara, Turkey), the University of Rochester (Rochester, NY), and Northwestern University (Evanston, IL), with courses spanning introductory physics laboratories to graduate-level quantum mechanics and particle physics. Developer of over 200 pages of supplementary course materials, written lab manuals, computational tutorials, and particle physics tools designed to deepen student engagement with the subject. Courses successfully adapted for remote instruction, with consistent emphasis on clarity, structure, and accessibility for students across levels, from undergraduates encountering computational methods for the first time to graduate students bridging adjacent subfields.

EDUCATION

Ph.D. in Physics
Northwestern University, Evanston, Illinois, USA
August 2020 – June 2025 (Conferred September 2025)
Dissertation: *Precision phenomenology and new physics probes at future colliders*
Advisor: Francis John Petriello

M.Sc. in Physics
Middle East Technical University, Ankara, Turkey
February 2017 – July 2019
Thesis: *Exploring extra dimensions through rare processes*
Advisor: İsmail Turan
Co-Advisor: İsmet Yurduşen (Hacettepe University)

B.Sc. in Physics (Double Major)
Middle East Technical University, Ankara, Turkey
September 2012 – February 2017
Thesis: *Exploring universal extra dimensions*
Advisor: İsmail Turan

B.Sc. in Civil Engineering
Middle East Technical University, Ankara, Turkey
September 2009 – February 2016
Thesis: *Redesign of METU pedestrian bridge*
Advisor: Alp Caner

EMPLOYMENT

U.S. Bank, Atlanta, Georgia, USA
Financial Data Scientist
August 2026 – Present

Kennesaw State University, Kennesaw, Georgia, USA
Postdoctoral Fellow in Nikolaos Kidonakis' group (University- and NSF-funded research)
August 2025 – August 2026

Northwestern University, Evanston, Illinois, USA
Teaching assistant (12 quarters)
Research assistant in Francis J. Petriello's group (4 quarters, DOE-funded research)
September 2020 – June 2025

Argonne National Laboratory, Lemont, Illinois, USA
Visiting student in Radja Boughezal's group (DOE-funded research)
January 2023 – July 2023

University of Rochester, Rochester, New York, USA
Teaching assistant
August 2019 – August 2020

Middle East Technical University, Ankara, Turkey
Teaching assistant
October 2017 – August 2019
Student assistant
October 2016 – June 2017

TEACHING INTERESTS

Quantum mechanics, particle physics, general physics.

RESEARCH INTERESTS

High-energy physics phenomenology: Precision in quantum chromodynamics and electroweak physics, collider physics, effective field theories, hadron physics, top physics, rare processes, theories with extra dimensions.

TEACHING EXPERIENCE

Teaching Assistant at Northwestern University
Undergraduate Level (7 quarters total)

- **PHYSICS 125-1** General Physics ISP *(1 qtr.)*: Graded homework and exam papers; designed discussion problems; led discussion sessions
- **PHYSICS 130-3** College Physics *(1 qtr.)*: Graded quiz papers; designed discussion problems; led discussion sessions
- **PHYSICS 135-2,3** General Physics: Electricity & Magnetism and Modern Physics *(1 qtr.)*: Graded quiz and exam papers; designed discussion problems; led discussion sessions
- **PHYSICS 136-2** General Physics Laboratory: Electricity & Magnetism *(1 qtr.)*: Led lab sessions; graded lab reports
- **PHYSICS 126-2,3** Physics for ISP Laboratory: Electricity & Magnetism *(2 qtrs.)*: Led lab sessions; graded lab reports

Graduate Level (9 quarters total)

- **PHYSICS 411-1** Methods of Theoretical Physics *(3 qtrs.)*: Graded homework and exam papers
- **PHYSICS 412-1,2,3** Quantum Mechanics *(5 qtrs.)*: Graded homework and exam papers; led discussion sessions; prepared 100+ pages of supplemental material including computational resources

- **PHYSICS 416-0** Introduction to Statistical Mechanics *(1 qtr.)*: Graded homework papers

Teaching Assistant at University of Rochester
Undergraduate Level (6 semesters total)

- **PHYSICS 113, 114** General Physics I & II Laboratory *(2 smtrs.)*: Designed lab manual; led lab sessions; graded lab reports
- **PHYSICS 121, 122** Mechanics and Electromagnetism Laboratory *(2 smtrs.)*: Designed lab manual; led lab sessions; graded lab reports
- **PHYSICS 142** Electricity & Magnetism Laboratory *(1 smtr.)*: Led lab sessions; graded lab reports
- **PHYSICS 123** Waves & Modern Physics *(1 smtr.)*: Led lab sessions; graded lab reports and homework and exam papers; led discussion sessions

Teaching Assistant at Middle East Technical University
Undergraduate Level (11 semesters total)

- **PHYS 105, 106** General Physics I & II Laboratory *(7 smtrs.)*: Led lab sessions; graded lab reports and quizzes
- **PHYS 207** Concepts of Modern Physics *(1 smtr.)*: Graded quiz papers
- **PHYS 407, 408** Particle Physics I & II *(3 smtrs.)*: Graded homework papers; led theoretical discussion sessions; designed quiz and homework problems; delivered lectures; taught shell scripting, Mathematica, FeynArts, FormCalc, Package X, LanHEP, and CalcHEP

Graduate Level (6 semesters total)

- **PHYS 507, 508** Quantum Mechanics I & II *(4 smtrs.)*: Graded homework and exam papers; led discussion sessions; prepared homework and exam problems; prepared 100+ pages of supplemental material; delivered lectures
- **PHYS 545, 546** Particle Physics I & II *(2 smtrs.)*: Graded homework and exam papers; design discussion and exam problems; delivered lectures

PUBLICATIONS

Total citations: 379, h-index: 9 (INSPIRE)

[17] N. Kidonakis, **K. Şimşek**, *Top quark differential distributions in the SMEFT*, [arXiv: 2610.30328].

[16] N. Kidonakis, **K. Şimşek**, *N^3LO soft-gluon corrections for $Z\gamma$ production*, [arXiv: 2608.12556].

[15] N. Kidonakis, **K. Şimşek**, *$t\bar{t}$ production as a probe of dimension-6 SMEFT at higher orders*, Phys. Rev. D **114** 055019 [arXiv: 2604.00902].

[14] N. Kidonakis, **K. Şimşek**, *Constraining dimension-6 SMEFT with higher-order predictions for $pp \to tW$*, Phys. Rev. D **113** (2026) 095004 [arXiv: 2601.21040].

[13] F. Petriello, **K. Şimşek**, *Naive T-odd Drell-Yan angular coefficients as a probe of the dimension-8 SMEFT*, Phys. Rev. D **113** (2026) 075029 [arXiv: 2511.19617].

[12] R. Boughezal, F. Petriello, **K. Şimşek**, *Transverse spin asymmetries and the electron Yukawa coupling at an FCC-ee*, Phys. Rev. D **110** (2024) 075026 [arXiv: 2407.12975].

[11] C. Bissolotti, R. Boughezal, **K. Şimşek**, *SMEFT analysis with LHeC, FCC-eh, and EIC DIS pseudodata*, [arXiv: 2307.09459].

[10] C. Bissolotti, R. Boughezal, **K. Şimşek**, *SMEFT probes in future precision DIS experiments*, Phys. Rev. D **108** (2023) 075007 [arXiv: 2306.05564].

[9] R. Boughezal, A. Emmert, T. Kutz, S. Mantry, M. Nycz, F. Petriello, **K. Şimşek**, D. Wiegand, X. Zheng, *Neutral-current electroweak physics and SMEFT studies at the EIC*, Phys. Rev. D **106** (2022) 016006 [arXiv: 2204.07557].

[8] R. Abdul Khalek et al. (including **K. Şimşek**), *Snowmass 2021 White Paper: Electron Ion Collider for High Energy Physics*, [arXiv: 2203.13199].

[7] T. M. Aliev, **K. Şimşek**, *Strong coupling constants of charmed and bottom mesons with light vector mesons in QCD sum rules*, Phys. Rev. D **104** (2021) 074034 [arXiv: 2107.02735].

[6] T. M. Aliev, T. Barakat, **K. Şimşek**, *Strong $B^*_{QQ'}B_{QQ'}V$ vertices and the radiative decays of $B^*_{QQ} \to B_{QQ}\gamma$ in the light-cone sum rules*, Eur. Phys. J. A **57** (2021) 160 [arXiv: 2101.10264].

[5] T. M. Aliev, **K. Şimşek**, *Strong vertices of doubly heavy spin-3/2–spin-1/2 baryons with light mesons in light-cone QCD sum rules*, Phys. Rev. D **103** (2021) 054044 [arXiv: 2011.07150].

[4] T. M. Aliev, T. Barakat, **K. Şimşek**, *Gravitational formfactors of the ρ, π, and K mesons in the light-cone QCD sum rules*, Phys. Rev. D **103** (2021) 054001 [arXiv: 2009.07926, arXiv: 2008.04385].

[3] T. M. Aliev, **K. Şimşek**, *Strong coupling constants of doubly heavy baryons with vector mesons in QCD*, Eur. Phys. J. C **80** (2020) 976 [arXiv: 2009.03464].

[2] H. I. Alrebdi, T. M. Aliev, **K. Şimşek**, *Determination of the strong vertices of doubly heavy baryons with pseudoscalar mesons in QCD*, Phys. Rev. D **102** (2020) 074007 [arXiv: 2008.05098].

[1] T. M. Aliev, T. Barakat, **K. Şimşek**, $N^*(1535) \to N$ *transition form-factors due to the axial current*, Phys. Rev. D **100** (2019) 054030 [arXiv: 1907.08017].

INVITED SEMINARS AND TALKS

[5] *Beyond-the-Standard-Model search at the Large Hadron-electron Collider and the Electron-Ion Collider*, invited talk at Physics and Astronomy Early Career Research Seminars, Northwestern University, April 19, 2023.

[4] *Precision electroweak measurements and beyond the Standard Model searches at the Electron-Ion Collider*, invited talk at DIS2023: XXX International Workshop on DIS and Related Subjects, Michigan State University, March 30, 2023.

[3] *SMEFT projections of neutral-current PVDIS asymmetries at the EIC*, invited talk at EIC Early Career Workshop 2022, Remote, July 25, 2022.

[2] *SMEFT projections using EIC PVDIS asymmetries*, invited talk at INT Workshop: Parity-Violation and other Electroweak Physics at Jlab 12 GeV and Beyond, Remote, June 27, 2022.

[1] *Neutral-current SMEFT studies at the EIC*, invited talk at CFNS Workshop: High-Luminosity EIC (EIC Phase II), Remote, June 21, 2022.

SEMINARS AND TALKS

[8] *Precision phenomenology and new physics probes at future colliders*, dissertation talk at Northwestern University, June 16, 2025.

[7] *SMEFT probes in future precision DIS experiments*, talk at New Perspectives 2023, Fermilab, June 27, 2023.

[6] *Precision electroweak measurements and SMEFT studies at the EIC*, talk at Phenomenology 2023 Symposium, University of Pittsburgh, May 9, 2023.

[5] *SMEFT projections at the EIC and LHeC to NLO QCD*, candidacy talk at Northwestern University, December 2, 2022.

[4] *Neutral-current SMEFT studies at the EIC*, seminar at HEP Seminars, Northwestern University, April 18, 2022.

[3] *Applications of MUED to rare top quark processes*, talk at 2020 GSRM Talks, University of Rochester, February 8, 2020.

[2] *Exploring extra dimensions through rare processes*, dissertation talk at Middle East Technical University, July 12, 2019.

[1] *Universal extra dimensions*, seminar at Middle East Technical University, December 6, 2018.

CONFERENCES, WORKSHOPS, AND SCHOOLS ATTENDED

[7] New Perspectives 2023, Fermilab, June 26–27, 2023.

[6] Phenomenology 2023 Symposium, University of Pittsburgh, May 8–10, 2023.

[5] DIS2023: XXX International Workshop on DIS and Related Subjects, Michigan State University, March 27–31, 2023.

[4] EIC User Group Early Career Workshop 2022, Remote, July 24–25, 2022.

[3] 2022 CTEQ Summer School on QCD and Electroweak Phenomenology, University of Pittsburgh, July 6–16, 2022.

[2] INT Workshop: Parity-Violation and other Electroweak Physics at Jlab 12 GeV and Beyond, Remote, June 27–July 1, 2022.

[1] CFNS Workshop: High-Luminosity EIC (EIC Phase II), Remote, June 21–24, 2022.

AWARDS

[4] Office of Research Postdoctoral Fellowship, Kennesaw State University, Aug 2025–Aug 2026.

[3] The most engaging and fun talk award, New Perspectives 2023, Fermilab, June 27, 2023.

[2] Scholarship covering full tuition and monthly stipend, Northwestern University, Aug 2020–Jun 2025.

[1] Scholarship covering full tuition and monthly stipend, University of Rochester, Aug 2019–Jun 2020.

COMPUTER SKILLS

Programming languages: Mathematica, Python, Fortran, R, C, C++, Matlab.
HEP software: FeynArts, FormCalc, LoopTools, FeynCalc, Package X, LanHEP, CalcHEP, LHAPDF, MadGraph, MCFM, MATRIX.
Data science and machine learning: SQL, Numpy, Pandas, Matplotlib, SciPy, Autograd, Pytorch, Tensorflow, Scikit-learn, unsupervised and supervised learning, neural networks, regression, classification, dimensional reduction, data visualization, hyperparameter tuning, cross-validation, synthetic dataset generation.
Statistical modeling and analysis: Maximum likelihood estimation, chi-square minimization, Bayesian inference, goodness-of-fit testing, multivariate regression, uncertainty quantification, hypothesis testing.
Scientific computing: High-performance cluster computing, MPI, batch job scheduling via Slurm/PBS, optimization, numerical simulations, Monte Carlo methods, large-scale data

processing.
Tools and platforms: Git/GitHub, Linux/Unix, shell scripting, TEX, Jupyter, Markdown, system administration.
Other skills: Algorithm design, data wrangling and cleaning, distributed computing workflows, pipeline automation, technical writing and documentation.

OTHER ACTIVITIES

Independent music composer and producer; releases available on Spotify and YouTube. Licensed chess player, Turkish Chess Federation.

LANGUAGES

Turkish (native), English (fluent), German (beginner), French (beginner), Spanish (beginner).

REFERENCES

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